

5. Constructor

- A constructor in Java is a block of code that initializes (constructs) the state and value during object creation.
- It is called every time an object with the help of a new () keyword is created.
- Even if you haven't specified any constructor in the code, the Java compiler calls a default constructor.
- The default constructor is used to assign default states and values, such as 0, null, etc., to the object.
- **A Java constructor cannot be abstract, static, final, and synchronized.**
- Syntax for default constructor is as follow

```
1 public class Animal{
2
3 public Animal(){
4
5 // default constructor
6 }
7
8 }
```

Key points -

- **Name:** It should have the same name as the class.
- Unique Features of Constructors in Java
- **No Return Type:** Unlike methods, a constructor has no return type.
- Automatically called: The constructor is automatically invoked when an object is created using the new keyword, relieving you from the need to call it explicitly.
- **Types:** There can be two types of constructors as follows:
- **Default Constructor:** The compiler automatically provides this constructor if no constructors are defined. This kind of constructor initializes objects with default values.
- **Parameterized Constructor:** A parameterized constructor allows objects to be initialized to specified values passed as arguments.

Parameterized Constructor

```
1 public class Animal{
2     String name;
3     int age;
4
5     // Parameterized Constructor
6     public Animal(String name, int age) {
7         this.name = name;
8         this.age = age;
9     }
10
11     void displayInfo() {
12         System.out.println("Name: " + name + ", Age: " + age);
13     }
14 }
15
16 public class Main {
17     public static void main(String[] args) {
18         // Creating an object of the Animal class
19         Animal a = new Animal("Tiger", 26);
20         a.displayInfo();
21     }
```

Extra miles → Visit link [Java Constructors](#) for more questions and details about constructor.

“ ***Confusion is part of programming.*** ” — Felienne Hermans

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